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ENTERPRISE NETWORK DESIGN FOR AFRIDATA SOLUTIONS LTD

Case Study-Based Assignment

TASK 1

The proposed architecture employs a robust hybrid topology, a structured IP addressing scheme, and dynamic routing to establish a reliable, secure, and scalable foundation for the company's operations across its Nairobi, Kisumu, and Mombasa branches.

The logical and physical topology selected for this design is a hybrid star-mesh configuration. Within each branch location, a star topology is implemented, where all end devices connect centrally to a layer-2 switch, which in turn connects to the local router. This approach simplifies local network management, troubleshooting, and containment of broadcast traffic.

For the Wide Area Network (WAN) interconnecting the three branches, a partial mesh topology is utilized. Each branch router is connected to the other two via dedicated serial links, creating redundant pathways. This design provides significant resilience; for instance, if the direct link between Nairobi and Kisumu fails, traffic can automatically reroute through Mombasa, thereby directly addressing the business challenge of frequent network downtime.

For transmission media, high-reliability serial leased lines are chosen for the WAN connections to simulate secure, point-to-point links, while cost-effective and high-performance Copper Ethernet is used for all Local Area Network (LAN) connections.

Wireless access is provided using IEEE 802.11n on the 2.4 GHz band, offering a balanced combination of range and throughput for office environments.

A critical component of the design is a hierarchical IP addressing scheme using Variable Length Subnet Masking (VLSM) to ensure efficient resource allocation. Each branch has been allocated a dedicated /24 network: Nairobi uses 192.168.1.0/24, Kisumu 192.168.2.0/24, and Mombasa 192.168.3.0/24. Within these networks, addresses are further segmented for specific functions. The router interfaces serving the LANs are assigned the first usable IP address (e.g., 192.168.1.1), servers are assigned static addresses in a reserved range (e.g., 192.168.1.100), and wireless access points are given static IPs for management (e.g., 192.168.1.50). The WAN links between routers use /30 subnets to conserve address space, with networks 192.168.10.0/30 for NBO-KSM, 192.168.20.0/30 for NBO-MSA, and

192.168.30.0/30 for KSM-MSA. This structured plan prevents IP address conflicts, simplifies administration, and supports future scalability.

To enable the interconnection between the LANs and WANs, the Open Shortest Path First (OSPF) dynamic routing protocol was implemented over alternatives like RIP or BGP. OSPF is chosen for its fast convergence, which minimizes downtime after a link failure, and its scalability, which accommodates future network expansion. All routers are configured within a single OSPF area (Area 0), and the protocol is enabled on all LAN and WAN interfaces. This configuration allows each router to dynamically learn the routes to all subnets across the enterprise. The successful establishment of OSPF neighbor adjacencies and the presence of remote network routes in the routing tables, verified via the *`show ip OSPF neighbor`* and `show ip route` commands, confirm that the routing is fully operational.

```
Router#show ip OSPF neighbo

Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           0   FULL/  -        00:00:37    192.168.20.2    Serial0/0/1
2.2.2.2           0   FULL/  -        00:00:33    192.168.10.2    Serial0/0/0
Router#
```

```
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
O    192.168.2.0/24 [110/65] via 192.168.10.2, 02:06:33, Serial0/0/0
O    192.168.3.0/24 [110/65] via 192.168.20.2, 02:06:33, Serial0/0/1
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/30 is directly connected, Serial0/0/0
L       192.168.10.1/32 is directly connected, Serial0/0/0
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/30 is directly connected, Serial0/0/1
L       192.168.20.1/32 is directly connected, Serial0/0/1
     192.168.30.0/30 is subnetted, 1 subnets
```

In conclusion, the network design presented provides a solid architectural foundation that directly meets AfriData's requirements. The hybrid star-mesh topology ensures reliability and ease of management, the comprehensive subnetting scheme allows for efficient IP resource allocation and growth, and the implementation of OSPF dynamic routing guarantees robust and self-healing connectivity between all branches. This design effectively mitigates the previously identified issues of downtime and poor performance, setting the stage for a modern, high-performing enterprise network.

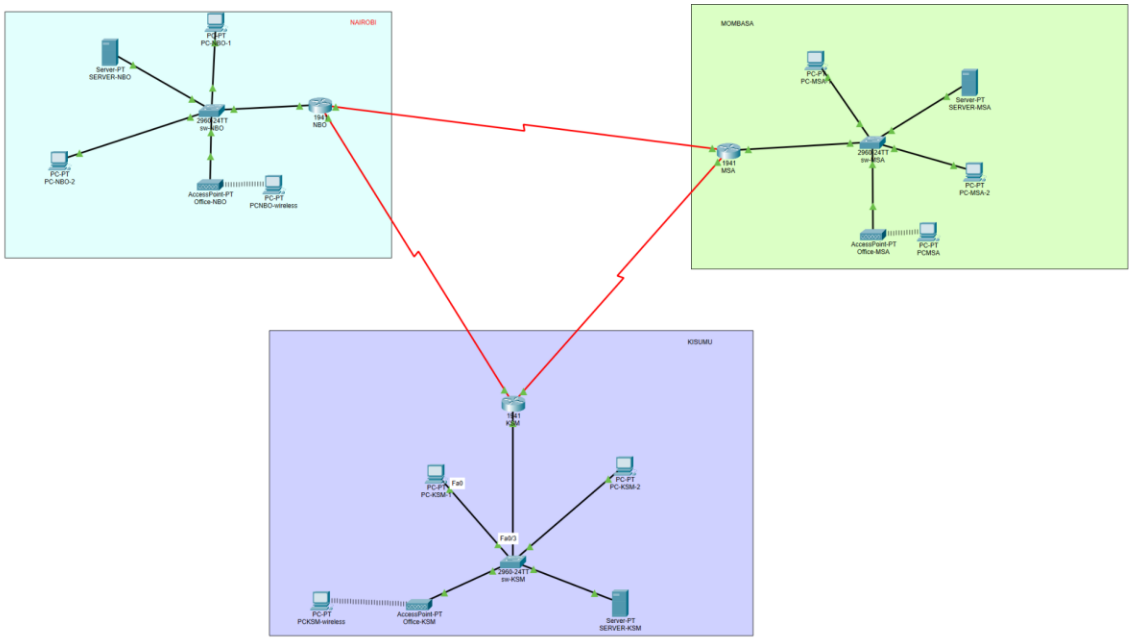
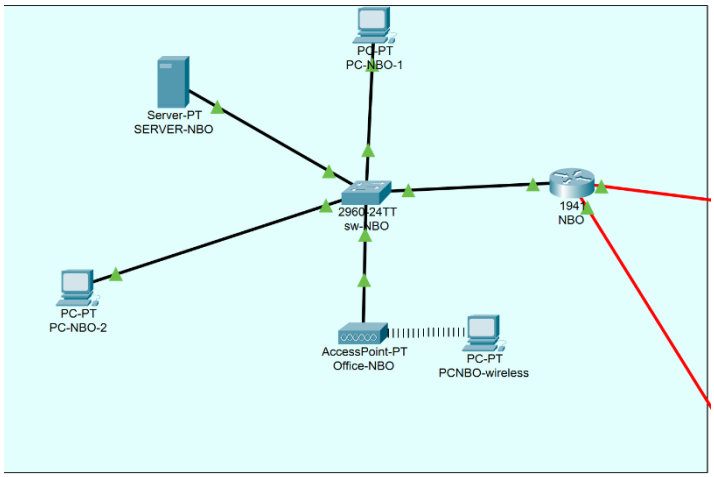
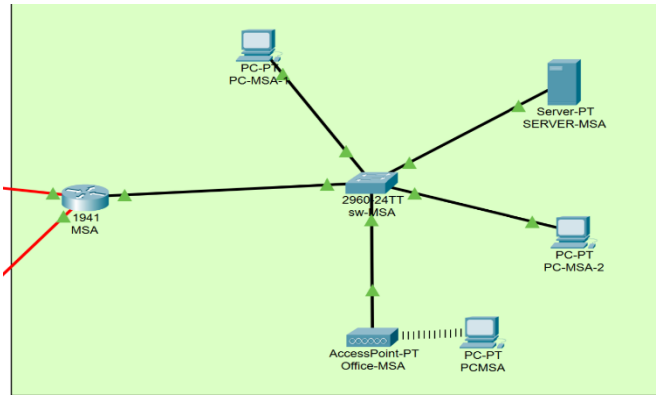
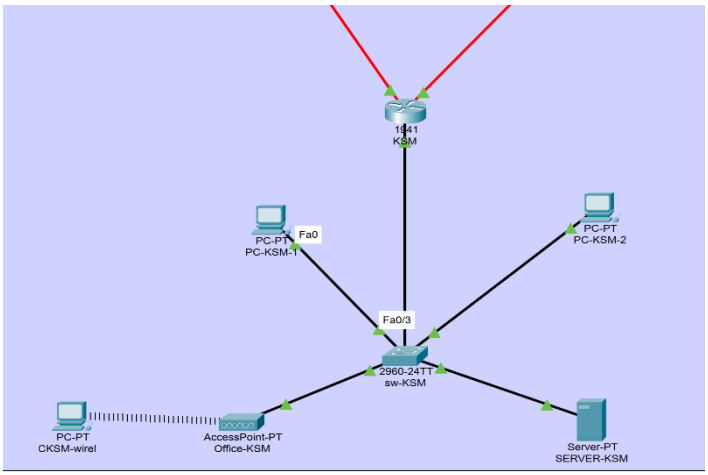


Figure 1 Diagram of the full physical network architecture







TASK 2

QUESTION 1:

The OSI and TCP/IP models will provide structured, layered frameworks for designing and implementing AfriData's network. They ensure compatibility, simplify troubleshooting and error detection, and guide protocol selection at each stage:

- The physical layer defines the electrical and physical specifications. This guides the physical topology, transmission mode, network interface cards, and hardware to ensure that the bit transmission between devices is reliable and efficient.

- The data link layer ensures effective node-to-node delivery by using MAC addresses to forward frames. Additionally, this layer checks for errors, and uses flow and access control mechanisms to prevent receiver bombardment and efficient communication. This guides the implementation by requiring the selected switches to efficiently handle these specific functions to maintain performance in a branch's network.

- The network layer manages logical addressing and routing across the different networks. It is crucial to the network's development because it handles the IP addressing scheme, and the choice of routing protocol (like OSPF) will determine the efficiency of data transmission across the AfriData network branches.

- The transport layer ensures reliable end-to-end communication, and its functions are critical in supporting AfriData's business operations: The segmentation and reassembly function breaks down data into segments for efficient and reliable transfer, while the service point function ensures that data for different services, for instance HTTP and SMTP, reach the correct applications. This guides the development by selecting protocols like TCP and UDP for guaranteed and fast delivery of data.

- The rest of the layers guide the software development and service configuration of the AfriData network: The session layer manages the communication dialogues, ensuring stable connection for video and audio services are properly established and terminated once the session ends. The presentation layer handles translation, encryption, and compression, which is important to maintain data security and to ensure information has the right format and can be read by the destination application. Finally, the application layer manages the end user

protocols and ensures that the underlying network design can support the business services provided by AfriData.

These models assist with the network development by breaking down the requirements for each of the layers, and determining the right technology and protocols that should be used to develop a network that is robust, scalable, and maintainable for AfriData's operations.

QUESTION 2:

DNS and DHCP fall under the application layer, and use UDP in the transport layer for fast communication. HTTP, FTP, and Email are also under the application layer, however they use TCP at the transport layer for reliable data transfer.

QUESTION 3:

Flow control is used to regulate the flow of data between different nodes in a network, it ensures stable communication by preventing buffer overflow (it is when a sending device transmits data at a faster rate than the receiving device can process, causing the receiver's temporary storage memory (buffer) to become full), it handles different data rates by ensuring the data flow matches the capacity of the receiving device, and uses network resources efficiently by avoiding packet loss and reducing the need for retransmissions.

On the other hand, congestion control is used to prevent congestion in a network, which is when too much data is being sent over a network, overloading the network and resulting in dropped packets and poor network performance. Congestion control is able to prevent this and enable stable communication in a network as it can sense congestion from packet loss, and upon detection of loss, the TCP will implement immediate cuts in the amount of data being transmitted by the sender. Congestion control will prevent collapse in data transfer by restricting

high-volume sending, and will boost efficient and stable communication by allowing the network to operate at its optimal capacity.

TASK 3

As a multi-site network, AfriData is vulnerable to a variety of threats considering it's fundamentally based on high connectivity and cloud usage. The top three threats to confidentiality, integrity, and availability of business are Distributed Denial of Service Attack

(DDoS), packet sniffing (public access to remote networks) and brute force attempts (unauthorized access) into the data stored on servers, clouds, VPN gateways, etc.

The best implemented security architecture relevant to AfriData would be the best-practice security architecture as it adequately focuses on perimeter, transmission and observational standards of security.

1. **Firewalls Security**: Each branch of AfriData should purchase Next Generation Firewalls (NGFW). They encompass application filtering, intrusion prevention and detection and can deny access due to access control. They are more advanced than simple packet filtering. For example, they can deny foreign IPs from accessing the network but also deny inbound access from domestic IPs attempting to gain access without any need of entry with unnecessary, irrelevant options. NGFW is superior to a firewall as it takes a more comprehensive approach towards security which an organization with so many branches' needs on a multi-tiered basis.

2. **Virtual Private Networks (VPN)**: VPNs are required. For inter-branch conversations, a site-to-site VPN will work fine. For persons out in the field who need information across branches or cloud access, client-to-site VPNs are encouraged. Tunneling will dissuade passive packet sniffing attempts across the public Internet as well as allow those needing to access data in network-based clouds and dashboards to more easily acquire information as their connection back to the internal private network is secure. In addition, applications solely reliant on the network for secure environments will be able to operate with greater clarity under a VPN with an encrypted connection versus unsecured access on a public network.

3. **Encryption Protocols (SSL/TLS)**: SSL/TLS encryption should be forced on internal web applications and dashboards within the internal email system. Therefore, for anyone accessing this information, they must seek HTTPS. Encryption protocols should also exist for inter-office emails and secure file transfer for sensitive analytical business data.

4. **Intrusion Detection and Prevention Systems (IDS/IPS)**: IDS/IPS should exist in the core of the network where sending packets are centralized, as it helps gather information for intelligence purposes. Packets are sent from the network core with IDS/IPS permission; therefore, access attempts of brute force can be diagnosed much sooner, patterns of IPs who do not belong (i.e. patterns of malware signatures) can be better understood and the visual

understanding of when problems occur and how they can be denied/intelligently denied beforehand is significant.

For example, an ACL for perimeter protection with firewalls was simulated in Packet Tracer:

access-list 101 deny tcp any any eq 23

access-list 101 permit tcp any any eq 443

interface GigabitEthernet0/1

ip access-group 101 in

Therefore, those who try to access unsecure Telnet information are denied, but secured users are granted access for HTTPS.

TASK 4

A complete network administration and performance management plan ensures that the network operates reliably, securely, and efficiently. The plan includes monitoring tools, performance metrics for evaluating the network's health, and a structured maintenance and escalation procedure.

Monitoring Tools:

Wireshark

- A packet-level network protocol analyzer.
- Used for deep inspection of traffic, packet tracing, identifying anomalies, diagnosing latency, and detecting security threats.
- Helps in analyzing TCP handshakes, retransmissions, duplicate ACKs, malformed packets, and unauthorized traffic.

SolarWinds Network Performance Monitor (NPM)

- Monitors device status, bandwidth utilization, network maps, and real-time alerts.
- Provides dashboards for throughput, SNMP device status, and link availability.
- Useful for detecting congestion and forecasting capacity requirements.

Key Metrics for Network Performance

To maintain high Quality of Service (QoS), the following metrics must be continuously monitored:

1. **Throughput**

- Measures actual data transfer rate across a link.
- Measured in Mbps or Gbps.
- Tools: SolarWinds, PRTG, iPerf.

2. **Latency**

- The **time a packet takes to travel** from source to destination.
- Measured in milliseconds (ms).
- High latency indicates congestion, poor routing, or overloaded devices.
- Tools: Ping, Wireshark (Time-Delta Between Packets).

3. **Availability**

- Percentage uptime of network devices and services.
- Formula:
- $Availability = \frac{Uptime}{Total\ Time} \times 100$
- Measured using SolarWinds or Nagios.

4. **QoS Metrics**

- Bandwidth allocation per class (Voice, Video, Data)
- Priority queues (e.g., weighted fair queuing)
- Traffic shaping and policing indicators

The network will be maintained regularly to prevent problems, and an escalation process will be followed whenever issues arise.

Regular Maintenance

- **Daily:** Check network health, device status, and any alerts.
- **Weekly:** Review logs and ensure backups and security updates are up to date.

- **Monthly:** Apply firmware/software updates and check for performance or security issues.
- **Quarterly:** Do a full network review and test disaster recovery procedures.

Escalation Procedure

- **Level 1:** Basic checks such as cable issues, rebooting devices, and simple connectivity tests.
- **Level 2:** A technician investigates deeper issues like configuration errors, DHCP/DNS problems, or VLAN issues.
- **Level 3:** A network administrator handles complex faults such as routing issues, firewall problems, or repeated outages.
- **Level 4:** If the issue is beyond internal control, it is escalated to the ISP or equipment vendor.

Sample Wireshark Output:

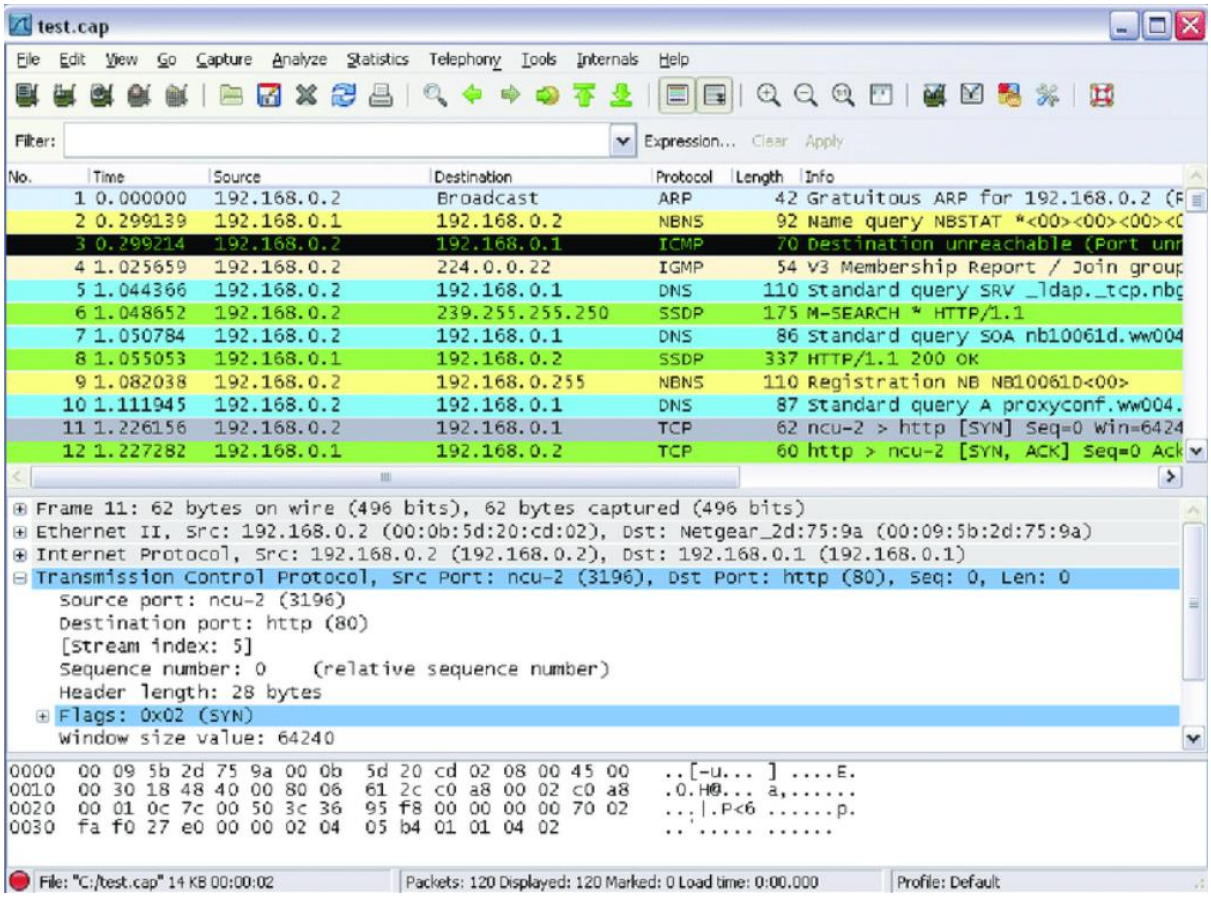


Figure 2 Research gate sample output (Shitharth & D, 2015)

This sample capture shows packet analysis as the Wireshark is capturing and breaking down every packet on the network, showing the source, destination, protocol, and detailed contents of each packet. A few abnormalities are also shown. There are ICMP "Destination Unreachable" messages, which mean that one device tried to reach another device or port, but it failed. There are also some ARP messages that could mean two devices might be trying to use the same IP address. The SSDP messages show devices on the network searching for other devices or services. Lastly, the TCP SYN packets show that a computer is trying to start a connection to a web server. All these help us see where the network might have small problems or misconfigurations.

TASK 5

Scalability Evaluation:

The proposed network is highly scalable as it utilizes a hybrid topology, which is designed to allow for easy integration of new hardware components. Additionally, the use of a hierarchical IP addressing scheme with VLSM can accommodate growth and any changes in network topology without requiring a complete redesign of the IP addressing scheme. The OSPF (Open Shortest Path First) protocol can segment the network into multiple areas, making it more scalable. However, scaling the WAN requires procuring and provisioning new physical leased lines for every new branch or link, thus increasing the cost and time needed for expansion, causing significant barriers to rapid growth. Manual configuration is needed when adding a new branch, which is prone to human error and is time-consuming, which might make it not very scalable.

Resilience Evaluation:

The partial mesh WAN topology provides redundancy, so if one link fails, traffic reroutes via another path, hence making it resilient. The use of hybrid topology allows for fault tolerance as it integrates multiple topologies, hence failure of a single node/segment does not result in a complete disaster for the entire network. Regular maintenance and escalation processes can help keep network health in check to prevent problems and recover quickly when issues arise. However, having a star topology means that all devices connect to a single device (layer 2 switch), so if this central switch fails, the entire local branch network will go down. The WAN

strategy is also solely dependent on one type of connection (leased lines) without any fallback option.

Security Posture Evaluation:

Next-Generation Firewalls (NGFW), VPNs, IDS/IPS, and SSL/TLS encryption are proposed, which provides a layered security. Monitoring tools such as Wireshark and SolarWinds help detect any security threats or anomalies. ACL and segmentation also help to prevent unauthorized access, while only secured users are granted access. However, reliance on perimeter security may not suffice for insider threats or compromised endpoints. Encryption is mostly focused on transmission, while data-at-rest encryption and cloud security controls are not emphasized.

Cloud Integration Upgrade:

To adopt a hybrid cloud model using a Cloud Adoption Framework (CAF), which can be used to migrate non-critical services to the hybrid cloud model to reduce on-premises load and improve scalability. It also provides good security and includes risk management.

SDN (Software Defined Networking) Upgrade:

To implement SD-WAN and SDA. SDA will allow the proposed network to move from manual configuration to an automated and simplified way of managing network access for users and devices. This is beneficial for managing access for a high number of devices. Furthermore, SDA will also be used to automate policy enforcement and provide granular micro-segmentation, which will drastically improve internal security. SD-WAN will increase resilience by providing dynamic, application-aware path selection across multiple connection types. It ensures optimal performance and provides automated failover, maintaining connectivity even if a primary link fails.

IoT Readiness Upgrade:

Implement an IoT Security Framework, including device authentication, specialized traffic monitoring, and edge computing capabilities to process data locally, reducing latency and WAN bandwidth consumption for IoT applications. The IoT security setup adds protections for all connected devices and helps monitor traffic close to where data is generated. Adding tools to

watch endpoints and bringing all security alerts into one place will make it easier to detect and fix problems more quickly.